Solution: Part I

Suppose there are N individuals, S states, and T periods. Each individual i belongs permanently to one state $s(i)$, and the panel is balanced.
Let

$$R \in \mathbb{R}^{N\times S}, \qquad R_{is} = \mathbf{1}\{s(i) = s\},$$

so each row of R contains exactly one 1, and

$$R'R = \operatorname{diag}(n_1, \ldots, n_S),$$

where n_s is the number of individuals in state s. Assume $n_s \geq 1$ for every state, so $R'R$ is invertible.
Let $Y \in \mathbb{R}^{N\times T}$ be the matrix with (i,t) entry Y_{it}, and let

$$\mathbf{D} \in \mathbb{R}^{S\times T}, \qquad \mathbf{D}_{st} = D_{st},$$

collect the state-time treatment values. Since treatment varies only at the state-time level,

$$D_{it} = D_{s(i)t},$$

the corresponding $N \times T$ matrix of individual-time treatment values is $R\mathbf{D}$.
We stack observations by individual first and then by time:

$$y := \operatorname{vec}(Y') \in \mathbb{R}^{NT}, \qquad d := \operatorname{vec}\big((R\mathbf{D})'\big) \in \mathbb{R}^{NT}.$$

Thus

$$y = (Y_{11}, \ldots, Y_{1T}, Y_{21}, \ldots, Y_{2T}, \ldots, Y_{N1}, \ldots, Y_{NT})'.$$

Define the dummy matrices

$$U := I_N \otimes \iota_T \in \mathbb{R}^{NT\times N}, \qquad \Delta := \iota_N \otimes I_T \in \mathbb{R}^{NT\times T}, \qquad W := R \otimes \iota_T \in \mathbb{R}^{NT\times S},$$

where ι_m denotes the $m \times 1$ vector of ones. Note that

$$W = UR.$$

Then the two regressions can be written as

$$y = W\alpha + \Delta\delta + \theta d + \varepsilon, \tag{1}$$

and

$$y = Uu + \Delta\delta + \theta d + \varepsilon. \tag{2}$$

We will show that the OLS estimate of θ is identical in these two regressions.

Projection notation. For any matrix A, let P_A denote the orthogonal projector onto $\mathrm{col}(A)$, and let

$$M_A := I - P_A.$$

Equivalently, one may think of $P_A = A(A'A)^+A'$, where $(\cdot)^+$ is the Moore–Penrose inverse. This avoids having to drop redundant dummy columns explicitly.

Step 1: Decompose the individual fixed effects into state means and within-state deviations. Define

$$P_R := R(R'R)^{-1}R', \qquad M_R := I_N - P_R.$$

For any $u \in \mathbb{R}^N$,

$$u = R\alpha + \eta, \qquad \alpha := (R'R)^{-1}R'u, \qquad \eta := M_R u.$$

Since $R'R = \mathrm{diag}(n_1, \ldots, n_S)$,

$$\alpha_s = \frac{1}{n_s} \sum_{i:s(i)=s} u_i, \qquad \eta_i = u_i - \alpha_{s(i)}.$$

Hence $R'\eta = 0$, i.e.

$$\sum_{i:s(i)=s} \eta_i = 0 \qquad \text{for every state } s.$$

Multiplying by U,

$$Uu = UR\alpha + U\eta = W\alpha + U\eta.$$

Equivalently,

$$U = W(R'R)^{-1}R' + UM_R.$$

Hence every column of U is the sum of a vector in $\mathrm{col}(W)$ and a vector in $\mathrm{col}(UM_R)$, so

$$\mathrm{col}(U) \subseteq \mathrm{col}(W) + \mathrm{col}(UM_R).$$

Conversely, since $W = UR$ and UM_R are both obtained by postmultiplying U,

$$\mathrm{col}(W) \subseteq \mathrm{col}(U), \qquad \mathrm{col}(UM_R) \subseteq \mathrm{col}(U),$$

and therefore

$$\mathrm{col}(W) + \mathrm{col}(UM_R) \subseteq \mathrm{col}(U).$$

Thus

$$\mathrm{col}(U) = \mathrm{col}(W) + \mathrm{col}(UM_R).$$

Step 2: The within-state individual deviations (UM_R) are orthogonal to state dummies (W) and time dummies (Δ). First,

$$W'UM_R = (UR)'UM_R = R'U'UM_R.$$

Since $U = I_N \otimes \iota_T$,

$$U'U = TI_N,$$

so

$$W'UM_R = T\,R'M_R = 0.$$

Next,

$$\Delta' UM_R = (\iota_N \otimes I_T)'(I_N \otimes \iota_T)M_R = \iota_T(\iota_N' M_R).$$

But $\iota_N = R\iota_S \in \text{col}(R)$, so $P_R\iota_N = \iota_N$, hence

$$\iota_N' M_R = \iota_N'(I_N - P_R) = 0.$$

Therefore

$$\Delta' UM_R = 0.$$

So $\text{col}(UM_R)$ is orthogonal to both $\text{col}(W)$ and $\text{col}(\Delta)$. It follows that

$$\text{col}([U, \Delta]) = \text{col}([W, \Delta]) \oplus \text{col}(UM_R),$$

Step 3: The treatment vector (d) is orthogonal to the within-state individual deviations (UM_R). Take any $a \in \mathbb{R}^N$. Then

$$Ua = (a_1\iota_T', \ldots, a_N\iota_T')'$$

is the vector that repeats a_i over time for each individual i. Hence

$$\begin{aligned} d'Ua &= \sum_{i=1}^{N}\sum_{t=1}^{T} D_{s(i)t}a_i \\ &= \sum_{s=1}^{S}\sum_{t=1}^{T} D_{st} \sum_{i:s(i)=s} a_i. \end{aligned}$$

If $a = M_R b$ for some b, then $R'a = 0$, which is equivalent to

$$\sum_{i:s(i)=s} a_i = 0 \qquad \text{for every state } s.$$

Therefore

$$d'UM_R b = 0 \qquad \text{for all } b \in \mathbb{R}^N,$$

so

$$d'UM_R = 0.$$

That is,

$$d \perp \text{col}(UM_R).$$

Step 4: The residualized treatment is the same under the two sets of fixed effects. Let

$$A_1 := [W, \Delta], \qquad A_2 := [U, \Delta].$$

From the orthogonal decomposition above,

$$\operatorname{col}(A_2) = \operatorname{col}(A_1) \oplus \operatorname{col}(UM_R).$$

Because $d \perp \operatorname{col}(UM_R)$, its projection onto $\operatorname{col}(A_2)$ is the same as its projection onto $\operatorname{col}(A_1)$:

$$P_{A_2} d = P_{A_1} d.$$

Hence

$$M_{A_2} d = M_{A_1} d.$$

Step 5: Apply Frisch–Waugh–Lovell. By the Frisch–Waugh–Lovell theorem, the coefficient on d in the state-FE regression is

$$\hat{\theta}_{\text{state}} = \frac{d' M_{A_1} y}{d' M_{A_1} d},$$

and the coefficient on d in the individual-FE regression is

$$\hat{\theta}_{\text{individual}} = \frac{d' M_{A_2} y}{d' M_{A_2} d}.$$

Since $M_{A_1} d = M_{A_2} d$ and M_{A_j} is symmetric,

$$d' M_{A_2} y = (M_{A_2} d)' y = (M_{A_1} d)' y = d' M_{A_1} y,$$

and likewise

$$d' M_{A_2} d = d' M_{A_1} d.$$

Therefore

$$\boxed{\hat{\theta}_{\text{individual}} = \hat{\theta}_{\text{state}}.}$$

Interpretation. Moving from state fixed effects to individual fixed effects adds only the time-invariant within-state deviations $UM_R u$. But the regressor d is constant within each state-time cell, so it is orthogonal to those extra directions. Therefore partialling out individual fixed effects instead of state fixed effects does not change the residualized treatment, and hence does not change the OLS coefficient on treatment.

Reference: DID

1. Basic DiD estimand

The population DiD estimand is

$$\theta_{\text{DiD}} = \Big(\mathbb{E}[Y_{i2} \mid G_i = 1] - \mathbb{E}[Y_{i1} \mid G_i = 1]\Big) - \Big(\mathbb{E}[Y_{i2} \mid G_i = 0] - \mathbb{E}[Y_{i1} \mid G_i = 0]\Big).$$

Its sample analog is

$$\widehat{\theta}_{\text{DiD}} = (\bar{Y}_{2,1} - \bar{Y}_{1,1}) - (\bar{Y}_{2,0} - \bar{Y}_{1,0}),$$

where $\bar{Y}_{t,g}$ is the sample mean of Y_{it} in period t and group $G_i = g$.
Using

	$t = 1$ (before)	$t = 2$ (after)
PA	6	5
NJ	10	7

we get

$$\widehat{\theta}_{\text{DiD}} = (7 - 10) - (5 - 6) = -3 - (-1) = -2.$$

2. Identification of the ATT

Start from

$$\begin{aligned}\theta_{\text{DiD}} &= \Big(\mathbb{E}[Y_{i2}(1) \mid G_i = 1] - \mathbb{E}[Y_{i1}(0) \mid G_i = 1]\Big) - \Big(\mathbb{E}[Y_{i2}(0) \mid G_i = 0] - \mathbb{E}[Y_{i1}(0) \mid G_i = 0]\Big) \\ &= \theta_0 + \Big(\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid G_i = 1] - \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid G_i = 0]\Big).\end{aligned}$$

Therefore,

$$\theta_{\text{DiD}} = \theta_0 \quad \Longleftrightarrow \quad \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid G_i = 1] = \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid G_i = 0].$$

So, given the timing assumption $Y_{i1} = Y_{i1}(0)$ for all i, the identifying condition is exactly *parallel trends in untreated potential outcomes*. It is both necessary and sufficient.

Interpretation. Absent the policy, average employment in NJ would have changed over time by the same amount as average employment in PA.

Why not directly testable? Because after the policy we never observe $Y_{i2}(0)$ for NJ stores. The object

$$\mathbb{E}[Y_{i2}(0) \mid G_i = 1]$$

is counterfactual.

3. Regression implementations of DiD

Let $\Delta Y_i = Y_{i2} - Y_{i1}$ and $\text{Post}_t = \mathbf{1}\{t = 2\}$.

(a) First-difference regression

Consider

$$\Delta Y_i = \alpha_0 + \theta_{\text{FD}} G_i + u_i, \qquad \mathbb{E}[u_i \mid G_i] = 0.$$

Then

$$\mathbb{E}[\Delta Y_i \mid G_i] = \alpha_0 + \theta_{\text{FD}} G_i.$$

Hence

$$\theta_{\text{FD}} = \mathbb{E}[\Delta Y_i \mid G_i = 1] - \mathbb{E}[\Delta Y_i \mid G_i = 0] = \theta_{\text{DiD}}.$$

(b) Saturated two-period DiD regression

Consider

$$Y_{it} = \beta_0 + \beta_1 G_i + \beta_2 \text{Post}_t + \theta_{\text{TW}} (G_i \cdot \text{Post}_t) + e_{it}, \qquad \mathbb{E}[e_{it} \mid G_i] = 0.$$

Because this is saturated in the four group-time cells,

$$\begin{aligned}
\beta_0 &= \mathbb{E}[Y_{i1} \mid G_i = 0], \\
\beta_1 &= \mathbb{E}[Y_{i1} \mid G_i = 1] - \mathbb{E}[Y_{i1} \mid G_i = 0], \\
\beta_2 &= \mathbb{E}[Y_{i2} \mid G_i = 0] - \mathbb{E}[Y_{i1} \mid G_i = 0], \\
\theta_{\text{TW}} &= \Big(\mathbb{E}[Y_{i2} \mid G_i = 1] - \mathbb{E}[Y_{i1} \mid G_i = 1]\Big) - \Big(\mathbb{E}[Y_{i2} \mid G_i = 0] - \mathbb{E}[Y_{i1} \mid G_i = 0]\Big).
\end{aligned}$$

Therefore

$$\theta_{\text{TW}} = \theta_{\text{DiD}}.$$

4. What trends are identified?

From the regression above,

$$\beta_2 = \mathbb{E}[Y_{i2} \mid G_i = 0] - \mathbb{E}[Y_{i1} \mid G_i = 0] = \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid G_i = 0].$$

Under parallel trends,

$$\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid G_i = 1] = \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid G_i = 0] = \beta_2.$$

So the average untreated trend is identified, and equals

$$\mathbb{E}[Y_{i2}(0) - Y_{i1}(0)] = \beta_2.$$

Now consider the policy trend for treated stores,

$$\mathbb{E}[Y_{i2}(1) - Y_{i1}(1) \mid G_i = 1].$$

This is *not* identified from the observed data without an extra restriction, because $Y_{i1}(1)$ is never observed. If we add a *no-anticipation* assumption,

$$Y_{i1}(1) = Y_{i1}(0) \quad \text{for treated stores,}$$

then

$$\begin{aligned}
\mathbb{E}[Y_{i2}(1) - Y_{i1}(1) \mid G_i = 1] &= \mathbb{E}[Y_{i2}(1) - Y_{i1}(0) \mid G_i = 1] \\
&= \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid G_i = 1] + \mathbb{E}[Y_{i2}(1) - Y_{i2}(0) \mid G_i = 1] \\
&= \beta_2 + \theta_0 = \beta_2 + \theta_{\text{TW}}.
\end{aligned}$$

5. Covariate-adjusted DiD

Consider

$$\Delta Y_i = \gamma_0 + X_i'\gamma + \theta_X G_i + v_i, \qquad \mathbb{E}[v_i \mid G_i, X_i] = 0.$$

A clean substantive set of conditions implying $\theta_X = \theta_0$ is:

(i) **Conditional parallel trends:**

$$\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid G_i = 1, X_i] = \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid G_i = 0, X_i] \quad \text{a.s.}$$

(ii) **Constant ATT given X_i:**

$$\mathbb{E}[Y_{i2}(1) - Y_{i2}(0) \mid G_i = 1, X_i] = \theta_0 \quad \text{a.s.}$$

Under these conditions,

$$\mathbb{E}[\Delta Y_i \mid G_i, X_i] = \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid G_i = 0, X_i] + \theta_0 G_i,$$

so the coefficient on G_i is θ_0.

Comparison with the baseline DiD assumption. This is weaker than unconditional parallel trends when differences in untreated trends across NJ and PA can be fully explained by observables X_i. It is not automatically weaker in every sense, however, because the regression imposes extra structure; in particular, with a single coefficient θ_X, it rules out systematic treatment-effect heterogeneity with respect to X_i.

6. Store closures and attrition

Let $R_i(d) \in \{0, 1\}$ indicate whether store i would remain open until period 2 under treatment state $d \in \{0, 1\}$. The complete-case DiD estimator conditions on observed $R_i = 1$. For it to identify the full-sample ATT θ_0, a strong sufficient set of assumptions is:

(i) **Treatment-invariant survival:**

$$R_i(1) = R_i(0) \quad \text{a.s.}$$

so treatment does not affect whether a store survives.

(ii) **Parallel untreated trends among survivors:**

$$\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid G_i = 1, R_i(0) = R_i(1) = 1] = \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid G_i = 0, R_i(0) = R_i(1) = 1].$$

(iii) **No survivor selection in treatment effects:**

$$\mathbb{E}[Y_{i2}(1) - Y_{i2}(0) \mid G_i = 1, R_i(0) = R_i(1) = 1] = \mathbb{E}[Y_{i2}(1) - Y_{i2}(0) \mid G_i = 1].$$

Under these assumptions, conditioning on $R_i = 1$ does not change the **target population** or the counterfactual trend comparison, so $\theta_{\text{CC}} = \theta_0$.

Evaluation. These assumptions are strong. The minimum wage may affect store survival, and the stores most likely to close are exactly those for which the treatment effect may be different.

A better strategy. If a closed store has zero employment in period 2, keep all stores from the baseline sample and set

$$Y_{i2} = 0 \quad \text{for stores that close.}$$

Then estimate DiD on the full baseline panel. This directly incorporates the extensive-margin effect of closures.

7. NJ-only before–after design

The NJ-only before–after estimand is

$$\theta_{\text{BA}} = \mathbb{E}[Y_{i2} \mid G_i = 1] - \mathbb{E}[Y_{i1} \mid G_i = 1].$$

Using potential outcomes,

$$\theta_{\text{BA}} = \mathbb{E}[Y_{i2}(1) - Y_{i1}(0) \mid G_i = 1] = \theta_0 + \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid G_i = 1].$$

Therefore

$$\theta_{\text{BA}} = \theta_0 \quad \Longleftrightarrow \quad \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid G_i = 1] = 0.$$

So the NJ-only design requires that, absent the policy, average NJ employment would not have changed at all over time. This is much stronger than DiD, which only requires NJ's untreated trend to match PA's untreated trend.

8. Alternative IV design

Suppose only PA stores are sampled. Let

$$Z_i = \mathbf{1}\{\text{store } i \text{ is offered the incentive}\}, \qquad D_i = \mathbf{1}\{\text{store } i \text{ adopts the higher minimum wage}\},$$

and let Y_i be post-policy employment.

(a) Wald/IV estimand

The IV estimand is

$$\theta_{\mathrm{IV}} = \frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0]}.$$

(b) LATE interpretation

Let $D_i(z)$ be adoption status under offer state $z \in \{0,1\}$, and let $Y_i(d)$ be the employment outcome under adoption state $d \in \{0,1\}$.
Under:

(i) **Random assignment:**

$$Z_i \perp (Y_i(1), Y_i(0), D_i(1), D_i(0));$$

(ii) **Exclusion:**

$$Y_i = Y_i(D_i),$$

so the offer affects employment only through adoption;

(iii) **First stage:**

$$\Pr(D_i = 1 \mid Z_i = 1) > \Pr(D_i = 1 \mid Z_i = 0);$$

(iv) **Monotonicity:**

$$D_i(1) \geq D_i(0) \quad \text{a.s.},$$

so there are no defiers;

the Wald estimand identifies

$$\theta_{\mathrm{IV}} = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i(1) > D_i(0)],$$

the LATE for compliers.

(c) Why it can differ from DiD/ATT

This IV estimand can differ from θ_0 for two reasons.
First, it is a *different population*: compliers in PA versus treated stores in NJ.
Second, it is a *different margin of variation*: voluntary adoption induced by an incentive versus a statewide policy change. With heterogeneous treatment effects, these need not coincide.